Ganesh Gupta

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Experience

Oracle | *Software Engineer, AI Platform* Jun 2025 – Present

- Provisioned and operated multi-cloud **GPU clusters** (AWS, GCP, CoreWeave) on **InfiniBand/RoCE** fabrics, tuning **transport-layer performance** for distributed training and inference at cluster scale.
- Wrote systems-level **scheduling, resource-allocation, and fault-tolerant placement** code in **Rust, Go, and C++** that keeps workloads running predictably under heavy churn and hardware scarcity.
- Built a pluggable **distributed executor** routing workloads across Docker and AWS Lambda with an **S3** artifact store at **sub-200ms p99**, debugging intermittent, **scale-sensitive failures**.
- Own production services end to end, from **design** through deployment, monitoring, and **on-call**, handling **millions of daily transactions** at **99%+ uptime**.

QureAI | *ML Infrastructure Engineer* Mar 2026 – Present

- Extended inference-serving internals with **kernel-level CUDA/Triton** development and a **distributed KV cache**, debugging performance-sensitive, intermittent failures.
- Cut **p99 latency from 420ms to 140ms** across **50K+ daily requests** via **performance tuning** and an async serving redesign.
- Ran **Prometheus/Grafana** observability and autoscaling for GPU serving, tracing latency regressions to **root cause**.

Nomura Research Institute Financial Technologies | *Software Engineer* Sep 2019 – Aug 2023

- Built and operated **25+ production C++ and Java** distributed services for **1M+ users** at **99.9% uptime** on AWS (EC2, RDS, SQS).
- Cut production escalations **60%** in one quarter by standardizing **retry, timeout, and failure-propagation** semantics across services.
- Cut nightly aggregation from **8 minutes to 90 seconds** by optimizing **SQL and PL/SQL** with indexed views and partitioning.
- Led **architecture and code reviews** across the full development lifecycle, mentoring engineers on **design patterns** and scaling.

Projects

VPC Control Plane Simulator / Distributed KV Store `github.com/ganeshhgupta/distributed-kv-store`

- **VPC control plane simulator** modeling **subnet routing**, traffic forwarding, and **security group enforcement**, alongside a **Raft**-consensus C++ store with linearizable reads and log compaction.

RootCause AI `github.com/ganeshhgupta/rootcause-ai`

- Multi-agent system that automates **production incident diagnosis**, tracing system-level failures across distributed services to a structured **root-cause** report.

Watch-your-LLM `github.com/ganeshhgupta/watch-your-llm`

- Observability platform tracking **latency** and regressions across model versions, with FastAPI, PostgreSQL, and **CI/CD**.

Education

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0* Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science* Aug 2016 – Jul 2020

Skills

Languages: C++, Rust, Go, Java, Python, SQL

Networking and Systems: InfiniBand/RoCE, Network Virtualization (VPC routing, security groups), Distributed Systems, Consensus Protocols, Performance Tuning

Infra: AWS (EC2, S3, Lambda, SQS), GPU Clusters, CUDA/Triton, Docker, Kubernetes, Prometheus, Grafana, CI/CD

Certifications: AWS Certified Developer Associate, GCP Professional MLOps Engineer